Output-Distribution Distillation for Contextual Sparsity Prediction in SwiGLU LLMs

Anonymous ACL submission

Abstract

Learned sparsity predictors for SwiGLU LLMs predominantly use binary cross-entropy (BCE), training each neuron independently, yet inference allocates budgets globally via top-k ranking. We train lightweight predictors (<1% of base parameters) with forward KL divergence, coupling all mask decisions through the output distribution. On LLaMA-3.1-8B at 30% sparsity, the KL predictor reduces perplexity by 14.5% over TEAL at matched sparsity, with directional commonsense gains (n=3) but knowledge-intensive regression (MMLU −5.2pp on 8B; GSM8K −21.2pp on 14B); post-hoc compensation (+67.2M params) becomes redundant. Output-distribution optimization yields masks that diverge 46–68% from magnitude-based targets; a target-swap ablation suggests these discovered patterns carry most of the quality information: BCE trained on KL-derived targets approaches KL-level PPL (PPL gap ≤0.65pp; 3-seed, 30%). Signal type (output-distribution vs. intermediate loss) may account for a substantial portion of the PPL gap at moderate sparsity (30%) (upper bound; §4.5). On downstream tasks, cross-layer coupling contributes substantially for commonsense reasoning but minimally for mathematical reasoning (3-seed K=4-corrected decomposition). Cross-architecture evaluation on Mistral-7B-v0.3 and Qwen-2.5-14B confirms PPL generalization with scale-dependent downstream effects. This is primarily an empirical study; wall-clock speedup requires sparse kernel support not addressed here.

1 Introduction

How the training objective shapes learned sparsity prediction quality is an open question for SwiGLU LLMs (Shazeer, 2020). SwiGLU FFN blocks exhibit high contextual sparsity, but their gated activation makes importance prediction challenging: neuron importance depends on both gate and up-projection values, a target that magnitude-based methods handle suboptimally. Learned sparsity predictors (Liu et al., 2023; Gautam et al., 2026) exploit this property by training small networks to anticipate which neurons are important, but they predominantly rely on binary cross-entropy (BCE) loss, treating each neuron as an independent classification target. Yet at inference, sparsity budgets are allocated globally: the TEAL (training-free activation sparsity; Liu et al., 2025) top-k mechanism ranks all neurons across all layers and retains only those above a model-wide threshold. This mismatch between per-neuron training and global inference allocation has not been formally analyzed. Unlike MaskLLM (Fang et al., 2024), which applies KL to static N:M weight masks, we address contextual activation sparsity where masks must generalize to unseen inputs, vary per token, and coordinate across layers through global allocation.

Gradient analysis reveals a structural asymmetry: BCE's gradient $\partial \mathcal{L}_{\text{BCE}}/\partial s_j = \sigma(s_j) - m_j^*$ depends only on neuron j, while under KL divergence, the gradient couples each neuron to all others through the downstream Jacobian (§3.6). KL's coupled gradients discover neuron importance patterns that diverge 68% from magnitude-based masks (mean per-layer Jaccard distance). A target-swap ablation shows KL-discovered mask targets carry most of the quality information: BCE retrained on KL-derived targets approaches KL PPL (≤0.65pp gap, 3-seed), while magnitude-trained BCE yields 28.13±0.20. Coupling contributes marginally to PPL but meaningfully to downstream fidelity (§4.5).

We instantiate a lightweight predictor (75.9M params, <1% of base) trained with forward KL and Gumbel-Sigmoid relaxation (Jang et al., 2017) on LLaMA-3.1-8B (Grattafiori et al., 2024). At 30% sparsity under uniform allocation, the predictor reduces TEAL's perplexity by 14.5% at matched sparsity (8.81 vs. 10.30, both at 28.5%) with downstream (MMLU 40.7±1.3% vs. 65.3% dense, 3-

seed); global top-k amplifies cross-seed variance ($7.3\times$). At 50%, the perplexity advantage widens (50% over TEAL), but downstream reasoning collapses regardless of method.

Cross-architecture evaluation on Mistral-7B-v0.3 (Jiang et al., 2023) (3-seed) confirms the PPL advantage (-17.3%) with directional commonsense gains but tied MMLU; preliminary evaluation on Qwen-2.5-14B (Team, 2024) shows consistent PPL improvement but reasoning decline (GSM8K -21.2pp, σ=6.8pp, n=2; MMLU -2.8pp). This paper is primarily an empirical study characterizing how the training objective shapes sparsity prediction quality across allocation strategies, sparsity levels, and model scales. The central finding—that output-distribution alignment consistently and substantially outperforms per-neuron classification for learned sparsity—is the paper's contribution. Without sparse kernel support, the predictor adds 6.4% latency overhead; wall-clock speedup is not addressed (Appendix A). While individual components (KL divergence, Gumbel-Sigmoid relaxation) are established techniques, their application to contextual activation sparsity prediction has not been explored, and the resulting empirical behaviors are non-obvious.

Our contributions are:

- *Empirical:* Output-distribution alignment ranking across five losses, with exploratory per-layer KL decomposition into signal type (may account for most of the PPL gap; upper bound) and coupling scope (task-dependent; K=4-corrected; §4.5);
- *Empirical:* Cross-scale evidence at 7–14B confirming output-distribution losses outperform intermediate losses regardless of coupling scope (§4).
- *Practical:* Target-swap ablation suggests KL-discovered mask targets transfer to BCE; coupling refines downstream fidelity beyond PPL (§4.5).
- *Practical:* These mask patterns render post-hoc compensation (+67.2M params) redundant at both tested sparsity levels (§4.5).

2 Related Work

Training-Free Activation Sparsity. TEAL (Liu et al., 2025) applies global top-k selection by activation magnitude and achieves 40–50% sparsity with no training cost. CATS (Lee et al., 2024) adapts per-layer thresholds for SiLU/SwiGLU activations; R-Sparse (Zhang et al., 2025b) predicts sparsity from rank-aware SVD projections; DIP (Federici et al., 2024) combines dynamic input pruning with cache-aware masking; SparseInfer (Shin et al., 2025b) and GRASP (Shin et al., 2025a) predict activation sparsity patterns without additional training. LaRoSA (Huang et al., 2025) applies layerwise orthogonal rotations for structured activation sparsity with wall-clock speedup on SiLU models. Wang et al. (Wang et al., 2025), AWQ (Lin et al., 2024), and Outlier Suppression+ (Wei et al., 2023) address sparsity–quantization interactions and activation-aware quantization. Dhar et al. (Dhar et al., 2024) characterize activation sparsity opportunities across architectures, showing that SwiGLU models exhibit lower natural sparsity than ReLU counterparts. OWL (Yin et al., 2024) determines per-layer sparsity ratios from outlier distributions for weight pruning. All training-free methods determine masks from fixed statistics or input-independent criteria, while our predictor generates *input-dependent* masks trained to minimize output-distribution divergence.

Training Signal Coupling →

Correction Granularity ↑

	Decomposed (BCE)	Intermediate (MSE)	Coupled (KL)
Per-token Contextual	*FastForward* *DejaVu*	—	★ **Ours**
Per-layer	—	*ShadowLLM*	—
Static	—	—	*SPON*

Figure 1: Method overview: a lightweight per-layer predictor produces importance scores trained via KL divergence with Gumbel-Sigmoid relaxation. At inference, scores feed TEAL's global top-k allocation without compensation networks.

Predictor-Based Contextual Sparsity. DejaVu (Liu et al., 2023) introduced learned sparsity predictors trained via BCE to classify each neuron independently. ShadowLLM (Akhauri et al., 2024) uses attention-based routing with per-layer MSE supervision for intermediate coupling within each layer but without end-to-end gradient signal. FastForward (Gautam et al., 2026) adds a 67.2M-parameter compensation network using BCE for mask prediction, adding inference latency. SVD-CSP (Serbin et al., 2026) compresses sparsity predictors via SVD decomposition for reduced predictor overhead. Dhar et al. (Dhar et al., 2025) compress activation patterns via clustering for efficient neuron-level prediction. LazyLLM (Fu et al., 2024) dynamically prunes tokens across layers for long-context efficiency; COTS (Zhang et al., 2025a) exploits connected activation patterns for structured sparse inference. PowerInfer (Song et al., 2024a) and PowerInfer-2 (Xue et al., 2024) exploit power-law activation frequencies for GPU–CPU/NPU scheduling. MaskLLM (Fang et al., 2024) shares the recipe-level principle of frozen weights + learnable masks + KL training. The methods differ in three respects: (a) MaskLLM's mask parameters are fixed per input (static N:M weight sparsity via Gumbel-Softmax), while our predictor generates *per-token* masks that must generalize to unseen inputs; (b) static N:M patterns have no cross-layer allocation interaction, whereas our masks coordinate through TEAL's global top-k; (c) activation masks must be robust to distribution shift at inference, while weight masks are input-independent. Our contribution beyond the shared recipe is analytical: a systematic decomposition of signal type vs. coupling scope for sparsity prediction training, via target-swap ablation and per-layer KL controls (§4.5). All prior predictor-based methods for activation sparsity use decomposed (BCE) or intermediate (MSE) objectives where each neuron's gradient is independent of other mask states. We empirically observe that an output-distribution-coupled objective propagates gradients through all downstream layers and quantify when this matters for global allocation quality.

Static Bias Correction. SPON (Spontaneous Neurons; Xu et al., 2025) learns input-independent bias vectors via closed-form KL minimization per layer, absorbed into weights at deployment. Our method is fundamentally different: a per-token contextual predictor trained end-to-end across all layers that produces input-dependent masks. SPON solves a static per-layer problem; we solve a global dynamic prediction problem. The two are weakly correlated (cosine similarity 0.196, R^2=0.038; §4).

Activation Function Engineering. ReLU Strikes Back (Mirzadeh et al., 2024) replaces SiLU with ReLU for >90% natural sparsity; TurboSparse (Song et al., 2024b) and ProSparse (Song et al., 2025) engineer extreme sparsity through activation modifications; the Sparsing Law (Luo et al., 2024) formalizes how activation density increases with training compute. MiniTron (Muralidharan et al., 2024) combines structured pruning with distillation but modifies architecture permanently. Haziza et al. (Haziza et al., 2025) demonstrate 2:4 semi-structured activation sparsity with hardware-native acceleration, complementing our unstructured approach. MoC (Wu et al., 2025b) selectively activates Top-K FFN channels using SwiGLU's native gating during pretraining. All require retraining or weight modification; our method is a plug-in predictor on *unmodified* SwiGLU models.

Weight Pruning, Distillation, and Routing. Wanda (Sun et al., 2024), SparseGPT (Frantar and Alistarh, 2023), and SliceGPT (Ashkboos et al., 2024) produce static weight masks via one-shot or structured pruning; Polar Sparsity (Shrestha et al., 2025) scales contextual sparsity to batched inference by shifting focus from MLP to attention head sparsity; Alizadeh et al. (Alizadeh et al., 2024) leverage activation sparsity for memory-efficient inference via selective flash loading. Switch Transformers (Fedus et al., 2022) and Mixture-of-Depths (Raposo et al., 2024) achieve conditional computation through learned routing. KL-based distillation (Kim and Rush, 2016; Sanh et al., 2019; Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024) trains compact students; SparseLoRA (Khaki et al., 2025) exploits contextual sparsity for fine-tuning. We apply the KL principle to train a sparsity predictor on unmodified dense models, using Gumbel-Sigmoid sampling (Maddison et al., 2017; Louizos et al., 2018) for differentiable discrete mask optimization.

MoE Routing and Gated Sparsity. Expert Choice routing (Zhou et al., 2022) lets each expert select its top-k tokens under a fixed capacity budget, a global allocation perspective that parallels TEAL's neuron-level global top-k. D2D-MoE (Szatkowski et al., 2024) converts dense SwiGLU layers into dynamic-k MoE by exploiting the same activation sparsity we target, but restructures the network into discrete experts rather than predicting masks on the original architecture. MoNE (Cheng et al., 2025) applies neuron-granular top-k selection within each MoE expert during pretraining for fine-grained conditional computation without additional routing parameters.

3 Method

We present an output-distribution KL framework for training contextual sparsity predictors in SwiGLU-based language models (Figure 1). The method replaces per-neuron binary classification (BCE) with an output-distribution alignment objective that couples all pruning decisions through the model's forward pass and achieves competitive performance without auxiliary error compensation networks at inference.

3.1 Problem Formulation

Consider a pretrained transformer with L layers, each containing a SwiGLU feed-forward network (Shazeer, 2020). For layer ℓ, the FFN computes:

$$\mathbf{y}^{(\ell)} = \left(\sigma(\mathbf{x}^{(\ell)}\mathbf{W}_g^{(\ell)}) \odot \mathbf{x}^{(\ell)}\mathbf{W}_u^{(\ell)}\right)\mathbf{W}_d^{(\ell)}, \quad (1)$$

where $\mathbf{x}^{(\ell)} \in \mathbb{R}^d$ is the hidden state after attention (d=4096), σ denotes SiLU, $\mathbf{W}_g^{(\ell)}, \mathbf{W}_u^{(\ell)} \in \mathbb{R}^{d\times d_{\text{ff}}}$ and $\mathbf{W}_d^{(\ell)} \in \mathbb{R}^{d_{\text{ff}}\times d}$ are the gate, up-projection, and down-projection matrices, and d_{ff}=14,336.

Define the gate activation $\mathbf{g}^{(\ell)} = \sigma(\mathbf{x}^{(\ell)}\mathbf{W}_g^{(\ell)}) \in \mathbb{R}^{d_{\text{ff}}}$. Each element $g_j^{(\ell)}$ determines the contribution of neuron j to the output. We introduce a binary mask $\mathbf{m}^{(\ell)} \in \{0,1\}^{d_{\text{ff}}}$ that zeros out pruned neurons:

$$\mathbf{y}_{\text{sparse}}^{(\ell)} = \left(\mathbf{m}^{(\ell)} \odot \mathbf{g}^{(\ell)} \odot \mathbf{x}^{(\ell)}\mathbf{W}_u^{(\ell)}\right)\mathbf{W}_d^{(\ell)}. \quad (2)$$

Our goal is to learn a mask predictor $f_\theta^{(\ell)}(\mathbf{x}^{(\ell)}) \to \mathbf{s}^{(\ell)} \in (0,1)^{d_{\text{ff}}}$ that produces per-neuron importance scores from $\mathbf{x}^{(\ell)}$. At inference, the scores are fed to a global top-k allocation mechanism (Liu et al., 2025) that distributes the sparsity budget across layers non-uniformly, retaining the $(1-\rho) \times L \times d_{\text{ff}}$ highest-scoring neurons model-wide for a target sparsity ratio ρ.

3.2 Predictor Architecture

Each layer ℓ has an independent two-layer MLP predictor:

$$f_\theta^{(\ell)}(\mathbf{x}^{(\ell)}) = \text{GELU}(\mathbf{x}^{(\ell)}\mathbf{W}_1^{(\ell)})\,\mathbf{W}_2^{(\ell)}, \quad (3)$$

where $\mathbf{W}_1^{(\ell)} \in \mathbb{R}^{d\times d_b}$ and $\mathbf{W}_2^{(\ell)} \in \mathbb{R}^{d_b\times d_{\text{ff}}}$ with bottleneck dimension d_b=128. The predictor takes $\mathbf{x}^{(\ell)}$ (the hidden state after attention, before the FFN) and outputs d_{ff}-dimensional logits $\mathbf{s}^{(\ell)}$, from which the mask is derived.

Each predictor has $(4096 \times 128) + (128 \times 14336) = 2.37$M parameters. With 32 independent predictors (one per layer), the total overhead is 75.9M parameters, less than 1% of the 8B base model. The predictors operate independently across layers, requiring no cross-layer communication at inference.

3.3 Gumbel-Sigmoid Relaxation

The hard mask $m_j = \mathbf{1}[s_j > \text{threshold}]$ is non-differentiable, preventing gradient flow from the KL objective through the mask to the predictor parameters. We apply Gumbel-Sigmoid relaxation (Maddison et al., 2017; Louizos et al., 2018; Jang et al., 2017) to produce differentiable soft masks:

$$\tilde{m}_j^{(\ell)} = \sigma_\tau\left(\log\frac{s_j^{(\ell)}}{1-s_j^{(\ell)}} + \varepsilon_1 - \varepsilon_2\right), \quad \varepsilon_1, \varepsilon_2 \sim \text{Gumbel}(0,1), \quad (4)$$

where $\sigma_\tau(x) = \text{sigmoid}(x/\tau)$ and τ is a temperature parameter. The Gumbel noise provides stochastic exploration of mask configurations during training, while the temperature controls sharpness: as $\tau \to 0$, $\tilde{m}_j$ approaches a binary decision.

We anneal τ linearly from 1.0 to 0.1 over training. At high temperature, the soft mask permits gradient flow through all neurons for broad exploration. As temperature decreases, the mask concentrates on confident decisions, approximating the hard allocation used at inference.

Comparison with Straight-Through Estimation. The straight-through estimator (STE) (Bengio et al., 2013) applies a hard threshold in the forward pass and passes gradients through unchanged ($\partial m/\partial s = 1$). In our setting, STE produces discontinuous gradient signals: small parameter changes can flip many neurons simultaneously. The resulting oscillations in the penalty weight prevent convergence to the target sparsity. Gumbel relaxation eliminates this pathology and contributes a 33.8% improvement (PPL 15.77 → 11.79 at 50% sparsity), as we verify empirically in §4.

3.4 KL Divergence Training Objective

Let $\mathbf{z}_{\text{dense}}$ denote the output logits from a full dense forward pass, and $\mathbf{z}_{\text{sparse}}$ the logits from a forward pass with predictor-generated soft masks $\{\tilde{\mathbf{m}}^{(\ell)}\}_\ell$ applied at every layer. We minimize the forward KL divergence between the dense and sparse output distributions:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(p_{\text{dense}} \,\|\, p_{\text{sparse}}) = \sum_v p_{\text{dense}}^{(v)} \log\frac{p_{\text{dense}}^{(v)}}{p_{\text{sparse}}^{(v)}}, \quad (5)$$

where $p_{\text{dense}} = \text{softmax}(\mathbf{z}_{\text{dense}})$ and $p_{\text{sparse}} = \text{softmax}(\mathbf{z}_{\text{sparse}})$, summed over the vocabulary v

and averaged over all token positions in the sequence. The base model is frozen; only the predictor parameters receive gradients.

We use forward KL (mass-covering) rather than reverse KL, as the goal is to preserve the dense model's primary predictions rather than explore alternative modes. In sparsity prediction, mode-covering is critical: false negatives (omitting a rarely but critically active neuron) cascade through downstream layers under global allocation, whereas false positives merely waste capacity; reverse KL's mode-seeking concentrates on high-probability neurons and risks missing compositionally important activations (Wu et al., 2025a). §4 validates this choice empirically: forward KL outperforms reverse KL by 8.6% and JSD by 6.4% (at 50% sparsity; the gap narrows at 30%; Table 4). No temperature scaling ($T{=}1$) is applied (Hinton et al., 2015); we distinguish T from the Gumbel temperature τ (§3.3), which controls mask sharpness rather than logit scaling.

3.5 Geometric Penalty Schedule

We enforce a model-level sparsity target ρ via constrained optimization with $\bar{s} = \frac{1}{L}\sum_{\ell}\frac{1}{d_{\mathrm{ff}}}\sum_{j}\tilde{m}_j^{(\ell)}$ denoting the average retained fraction:

$$\min_{\theta}\ \mathcal{L}_{\mathrm{KL}} \quad \text{s.t.} \quad \bar{s} = 1-\rho. \tag{6}$$

We relax this via a geometric penalty schedule:

$$\mathcal{L} = \mathcal{L}_{\mathrm{KL}} + \lambda \cdot \mathrm{clamp}(|\bar{s}-(1-\rho)| - \epsilon, \min{=}0)^2, \tag{7}$$

where $\epsilon = 0.02$ is a deadzone margin that suppresses oscillation near the target (sensitivity in Appendix C). The penalty weight λ is updated via geometric doubling (doubled when sparsity exceeds target, halved otherwise), clamped to $[\lambda_{\mathrm{floor}}, \lambda_{\mathrm{max}}]$, reaching target sparsity within 40–110 steps across all tested regimes (Appendix C).

3.6 Gradient Analysis: BCE vs. KL

We contrast the gradient structure of BCE and KL to build intuition for how KL discovers divergent mask patterns (§4.4). This analysis is suggestive; empirical evidence in §4.4 shows KL-discovered mask targets drive inference quality.

Under BCE, the predictor trains against oracle masks $m_j^* = \mathbf{1}[|g_j| \geq \text{top-}k \text{ threshold}]$ derived from activation magnitudes. The gradient $\partial\mathcal{L}_{\mathrm{BCE}}/\partial s_j = \sigma(s_j) - m_j^*$ depends only on neuron j; each neuron is trained independently with no inter-neuron coupling. Under output-distribution KL, the chain rule introduces a downstream Jacobian $\partial\mathbf{z}_{\mathrm{sparse}}/\partial\tilde{m}_j^{(\ell)}$ that couples neuron j's gradient to all other retained neurons in layers ℓ through L, because the sparse hidden state propagates through subsequent attention and FFN operations. This coupling allows KL to evaluate each neuron's importance in the context of the full sparse computation, rather than against a fixed magnitude target.

Table 1: Taxonomy of learned sparsity prediction methods along correction granularity (rows) and training signal coupling (columns). Our method occupies the Per-token × Coupled cell (organizational framework). Empty cells represent unexplored combinations; their relative merit is an open question.

	Decomposed (BCE)	Intermediate (MSE)	Coupled (KL)
Static	—	—	SPON, MaskLLM
Per-token	DejaVu, FastForward	ShadowLLM, D2DMoE	**Ours**

Gradient SNR measurements[1] reveal that BCE maintains consistently higher SNR than KL throughout training (2.3× at convergence; Figure 6), indicating that KL's quality advantage does not arise from cleaner gradient signals. Rather, KL's end-to-end coupled gradient serves as a *search mechanism* that discovers compositionally superior mask patterns, even when the per-step signal is noisier than BCE's decomposed updates. The target-swap ablation in §4.4 suggests that the discovered mask targets, not the coupling mechanism itself, drive inference-stage quality.

3.7 Two-Dimensional Taxonomy of Sparsity Prediction Methods

We organize prior work along two axes: *correction granularity* (how the mask adapts to input) and *training signal coupling* (whether the objective captures inter-neuron dependencies).

Our method combines per-token prediction with coupled KL training (Table 1). KL on output logits directly optimizes next-token prediction loss, which composes across layers; per-layer MSE does not—a mask minimizing FFN output error at layer ℓ may amplify errors in layers $\ell{+}1 \ldots L$, explaining its $1.7\times$/$9.6\times$ worse PPL than TEAL at 30%/50% (50%: s42 only; §4). Full taxonomy discussion in Appendix O.

[1] $\mathrm{SNR}_\ell = \|\mathbb{E}[\nabla_{\theta_\ell}\mathcal{L}]\|_2^2/\mathrm{Var}[\nabla_{\theta_\ell}\mathcal{L}]$, expectation over mini-batches.

4 Experiments

4.1 Experimental Setup

Model and Predictor. We evaluate on LLaMA-3.1-8B (Grattafiori et al., 2024), which uses SwiGLU FFN blocks (Shazeer, 2020) with $d_{\text{model}} = 4096$ and $d_{\text{ff}} = 14336$ across 32 layers. The sparsity predictor is a two-layer MLP with 2.37M parameters per layer (75.9M total), implemented as Linear(4096 $\rightarrow$ 128) + GELU + Linear(128 $\rightarrow$ 14336), trained to produce per-neuron importance scores that minimize output-distribution divergence under sparsity.

Training. All predictors are trained for 1000 steps on WikiText-103 with sequence length 2048 and batch size 4 on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB), using AdamW (Loshchilov and Hutter, 2019) with learning rate 1×10^{-3}, weight decay 1×10^{-2}, and cosine decay. The KL predictor uses Gumbel-Sigmoid relaxation (Jang et al., 2017) with temperature $\tau = 1.0$ annealed to 0.1 over training.

Evaluation. We report perplexity on WikiText-2 (Merity et al., 2017) (seq_len=2048) and downstream accuracy on ARC-Challenge (25-shot), WinoGrande (5-shot), MMLU (5-shot), GSM8K (8-shot), HellaSwag (10-shot), and PIQA (0-shot).

Inference Protocol. At inference, the predictor produces per-neuron scores that are passed to TEAL's global top-k allocation mechanism (Liu et al., 2025), which distributes the sparsity budget across layers non-uniformly. We term this *predictor-TEAL*: the predictor replaces TEAL's magnitude-based scoring while retaining its allocation strategy.

Baselines. We compare against: (1) **Vanilla TEAL** (Liu et al., 2025), which uses activation magnitudes directly[2]; (2) **BCE predictor** (DejaVu-style (Liu et al., 2023)), trained with per-neuron binary cross-entropy on magnitude-based oracle masks for 1000 and 2700 steps; and (3) **Uniform allocation**, which distributes the sparsity budget equally across layers.

4.2 Main Results

Table 2 presents the primary comparison. Uniform allocation preserves downstream performance best; global allocation is analyzed to understand allocation-quality tradeoffs (§4.6). At 30% sparsity, output-distribution KL under uniform allocation achieves PPL 8.81$\pm$0.02 (3-seed), a 14.5% reduction over TEAL at matched actual sparsity (8.81 vs. 10.30, both at 28.5%; Appendix D). Uniform allocation yields stable downstream (MMLU 40.7$\pm$1.3%, 3-seed; Appendix F), while global top-k exhibits high seed variance (35.7$\pm$9.5%) from late-layer starvation (L31 retention 28.4%; §4.6).

Task-Type Split. KL shows directional improvements over TEAL on commonsense reasoning (ARC-c, WG, HellaSwag, PIQA: 4/4 wins at 30% uniform) but underperforms on knowledge-intensive tasks (MMLU, GSM8K: 0/2). With only three seeds, 95% confidence intervals for individual downstream tasks overlap; we therefore treat PPL (σ=0.02, non-overlapping) as the primary metric and downstream differences as directional evidence. The KL predictor protects L14–L18 (14–18% sparsity vs. 32% global mean) while aggressively pruning L22–L28 (48–57%). Constraint interventions (Appendix T) confirm a task-dependent trade-off: removing the valley improves MMLU +3.4pp but reduces ARC-c by 1.5pp, consistent with localized factual associations (Geva et al., 2023); task-aware allocation may resolve this.

Knowledge-Task Regression: Hypothesis. We consider two hypotheses. First, WikiText-103's limited mathematical content may bias allocation against reasoning pathways—however, a C4-trained control does not support this (see below). Second, forward KL's mode-covering property distributes mass across high-frequency patterns, potentially diluting rare activations encoding factual knowledge concentrated in specific FFN neurons (Meng et al., 2022). Constraint experiments (Appendix T) confirm allocation contributes: removing the learned valley improves MMLU +3.4pp and GSM8K +1.6pp. A predictor trained on C4 (diverse web text; Appendix V) does not alleviate the regression (MMLU 40.0%, GSM8K 4.1% vs. 40.7%, 7.5%), confirming the deficit is structural to output-distribution KL rather than training data composition.

At 50%, the PPL advantage widens (11.92 vs. 23.90, 50% reduction) and generalizes to C4 (Appendix D), but all methods collapse to near-random on MMLU and GSM8K ($\leq$2.7%); no allocation strategy recovers at this capacity limit.

[2] TEAL is deterministic; single-run evaluation suffices. KL/BCE predictors require multi-seed evaluation due to random initialization.

Table 2: Main results on LLaMA-3.1-8B. *2.71× KL compute (§4.3). ±: 3-seed; unmarked: s42. ‡Actual sparsity: KL 28.5% vs. TEAL 29.3% at 30% (Appendix D).

Sparsity	Method	PPL↓	ARC-c	WG	MMLU	GSM8K
0%	Dense	6.12	58.28	78.06	65.27	48.75
30%	TEAL uniform	10.82	39.33	65.04	**45.86**	**8.79**
	BCE global	11.25	41.2	66.0	36.3	6.0
	BCE uniform	15.09±0.13	33.45	61.09	43.05	3.97
	MSE uniform	18.52±1.68	35.8±4.1	59.8±2.6	30.7±2.6	2.3±0.7
	KL global (Ours)‡	8.96±0.02	**43.8±3.6**	**70.2±0.9**	35.7±9.5	8.0±0.7
	KL uniform‡	**8.81±0.02**	43.7±2.5	68.9±0.4	40.7±1.3	7.5±0.1
50%	TEAL uniform	23.90	27.47	54.38	**29.26**	1.74
	BCE (2700 steps)*	28.13±0.20	27.6±0.6	51.8±0.6	26.1±0.5	2.1±0.2
	MSE uniform	229.26	25.00	50.59	23.79	1.59
	KL global (Ours)	**11.84±0.15**	**36.3±3.2**	**63.8±0.3**	28.3±0.7	**2.7±1.3**
	KL uniform	11.92±0.07	33.8±1.2	62.9±0.1	28.0±0.6	2.2±0.1

Table 3: Target-swap ablation (C1): BCE on KL-derived masks vs. KL predictor (same seed), TEAL uniform. All seeds show max $\Delta \leq 1.70$pp at 30%; max $\Delta \leq 0.79$pp at 50% (s42). †**Provenance:** s42 oracle extracted from an earlier training run (not seed-matched); s123/s456 use seed-matched retrained oracles (Appendix P).

Sparsity	Seed	Method	PPL↓	ARC-c	WG	MMLU	GSM8K
30%	s42†	C1	9.45	40.27	67.40	**42.30**	**7.58**
		KL	**8.82**	**40.78**	**68.59**	40.6	7.35
	s123	C1	8.84	44.71	**69.77**	**39.92**	6.82
		KL	**8.83**	**45.31**	69.38	39.4	**7.51**
	s456	C1	8.82	44.71	**69.93**	41.04	6.60
		KL	**8.79**	**45.05**	68.75	**42.0**	**7.58**
50%	s42	C1	11.96	32.17	**63.14**	27.79	**2.05**
		KL	11.96	**32.59**	62.35	**27.92**	1.97

Table 4: Ablation on LLaMA-3.1-8B (WikiText-2 PPL↓; uniform at 30%, global at 50%). ±: 3-seed; unmarked: s42. ‡STE: 20.8% actual sparsity at 30% target; loss effect (ΔPPL=20.4) is 5.1× relaxation effect (ΔPPL=3.98). §MSE 50%: s42 only; s123 catastrophic (PPL=15,789; Appendix G).

Configuration	30% PPL↓	50% PPL↓	Δ_{50}
KL-based variants			
Forward KL + Gumbel (Ours)	**8.81±0.02**	**11.79**	—
JSD (symmetric) + Gumbel	8.90±0.02	12.54	+0.75
Reverse KL + Gumbel	9.26±0.07	12.80	+1.01
Forward KL + STE	—‡	15.77	+3.98
Per-layer training			
Per-layer MSE	18.52±1.68	229.26§	+217.47
Per-layer KL (K=1)	9.05±0.02	34.85	+23.06
Per-layer KL (K=4)	8.89±0.01	44.23	+32.44
BCE-based variants			
BCE (no compensation)	15.71	27.97	+16.18
BCE + Gumbel (no penalty)	—	32.22	+20.43
BCE + Gumbel + Penalty	—	38.02	+26.23
BCE + STE	—	60.54	+48.75

4.3 Extended Training Analysis

At 2.71× KL compute budget (2700 steps, matched wall-clock), BCE achieves PPL 28.13±0.20 (3-seed), still 2.39× worse than KL (11.79). The gap is systematic: BCE reaches higher oracle F1 (0.613 vs. 0.540) with 139% worse PPL, confirming KL optimizes for distributional fidelity rather than magnitude agreement.

4.4 Mask Pattern Analysis

KL-derived masks diverge from magnitude oracles: per-layer Jaccard distance averages 68% at 50% (64 samples, 131K tokens; Figure 8), with cross-seed variation (36.3%) confirming systematic divergence (Appendices F.1, M). Despite this, KL achieves PPL 11.79 vs. BCE's 28.13±0.20, confirming that distributional fidelity, not magnitude agreement, produces better masks. Divergence follows a U-shaped per-layer pattern, peaking in late layers (0.70–0.75 at 50%).

Target-Swap Ablation: Target Quality vs. Gradient Coupling. To disentangle whether KL's advantage stems from its coupled gradient pathway or from the mask patterns it discovers, we train a standard BCE predictor on KL-derived oracle masks instead of magnitude-based masks (C1 target-swap ablation). Table 3 reports results under TEAL uniform allocation at matched sparsity.

At 50% (s42, earlier oracle), C1 matches KL within 0.79pp, while magnitude-trained BCE yields 2.4× worse PPL (Table 2). At 30%, retrained seeds show ≤1.18pp agreement[3]. C1 demonstrates that KL-discovered mask patterns *transfer* to a simpler training mechanism (PPL within 0.79pp; downstream variation up to 4.5pp MMLU); per-layer KL controls (§4.5) further show coupling contributes marginally to PPL but substantially to downstream fidelity. However, C1 also changes mask representation: binarizing oracle soft logits (>0) into hard masks discards ranking information, so the near-zero delta reflects transferability of binary mask *patterns* rather than full predictor knowledge. BCE+Gumbel (32.22) vs. standard BCE (27.97) suggests softness alone does not explain KL's advantage, though trajectory-dependent confounds remain (Appendix P).

4.5 Ablation Studies

Table 4 isolates the contribution of each design choice at 30% and 50% sparsity.

Relaxation, Divergence, and SPON. At 50%, Gumbel-Sigmoid outperforms STE (Bengio et al., 2013) by 33.8% (11.79 vs. 15.77); forward KL outperforms JSD (+6.4%) and reverse KL (+8.6%), consistent with mode-covering (Gu et al., 2024). At

[3]s42 oracle from earlier checkpoint (gap 0.65pp); s123/s456 use seed-matched converged oracles (≤0.03pp gap).

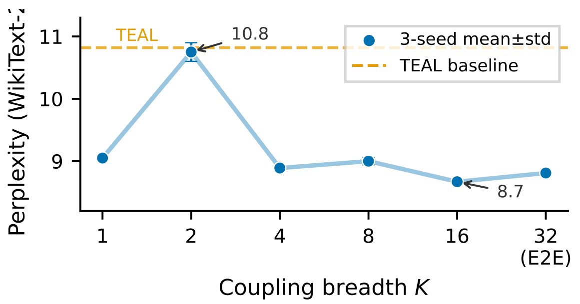

Figure 2: Coupling breadth K vs. perplexity (30%, TEAL uniform, 3-seed mean±std). K=32 denotes end-to-end training. K=2 degradation reflects insufficient per-layer updates (∼62 vs. ∼250 at K=8).

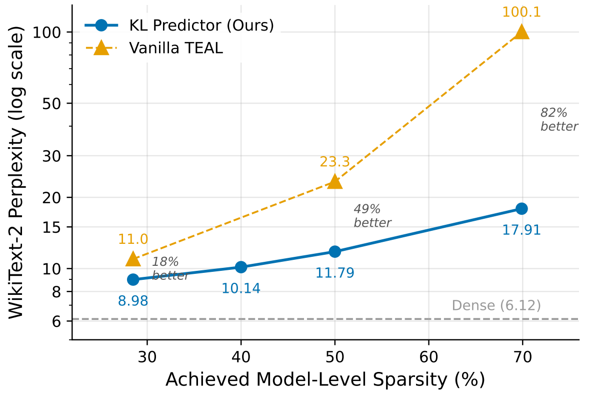

Figure 3: PPL scaling with sparsity (global allocation). KL's advantage over TEAL grows with sparsity.

30%, forward KL (8.81) and JSD (8.90) are near-identical, both outperforming reverse KL (9.26) and MSE (18.52±1.68). KL–JSD near-equivalence suggests the divergence measure matters less than optimizing through the output distribution; coupling scope provides additional downstream benefits (§4.5).

Per-Layer KL: Exploratory Decomposition of Signal Type and Coupling Scope. To isolate cross-layer coupling, we train per-layer KL predictors that mask K random layers per step and compute KL on the full model output, preserving the output-distribution signal without cross-layer gradients. At K=1, per-layer KL achieves 9.05±0.02, within 2.7% of end-to-end KL (8.81±0.02), while per-layer MSE degrades to 18.52±1.68 (2.05× worse). For PPL, signal type[4] may account for a substantial portion of the gap—an upper bound, as K=1 conflates coupling removal with train–inference mismatch. A 3-seed K=4 control[5] partially deconfounds this (Appendix S): *coupling-dominated*—ARC-c (97% residual gap) and WG (67%) indicate genuine cross-layer benefits; *mismatch-dominated*—PPL (67% closure) and GSM8K (71%) largely close; *anomalous*—MMLU exceeds E2E (42.7±1.4 vs. 40.7±1.3), explained by K=4's near-uniform allocation matching uniform inference (Appendix T). At 50%, partial coupling (K=4: 44.23) is *worse* than K=1 (34.85) while E2E achieves 11.84, suggesting gradient interference under tight capacity; the decomposition is interpretable only at 30%.

[4]We use "signal type" as shorthand for the training objective's optimization target (output distribution vs. intermediate activations); this dimension also entails differences in gradient flow topology and optimization landscape that are not independently controlled.

[5]Actual sparsity 29.6%; per-layer sparsity std=0.003 (near-uniform).

The K-curve (Figure 2) confirms non-monotonic coupling effects: K=16 achieves the lowest PPL (8.67±0.02 vs. E2E 8.81±0.02), yet the full coupling range (2.08 PPL) is dwarfed by the signal type gap (9.47 PPL).

Compensation Redundancy. FastForward-style compensation (Gautam et al., 2026) (+67.2M params) provides no benefit (Appendices J–B.2).

4.6 Sparsity Scaling and Regime Analysis

Allocation Behavior. Table 14 (Appendix L) reveals non-monotonic allocation preference: uniform is best at ≤30% (7.3× smaller MMLU σ), global at 35–40%, both collapse at ≥45%.

Cross-Architecture and Scale. On Mistral-7B-v0.3 (Jiang et al., 2023), PPL generalizes (−17.3%, 3-seed; Table 17); on Qwen-2.5-14B (Team, 2024), PPL generalizes but MMLU/GSM8K decline (Table 13). The consistent regression across scales (8B MMLU −5.2pp, 14B GSM8K 30.9±6.8 vs. 52.1, n=2) suggests a systematic property of output-distribution KL; Qwen's stronger reasoning baseline (GSM8K 87.6% vs. 48.8% dense) may amplify sensitivity to distribution-based pruning. The pipeline adds 6.4% latency overhead absent sparse kernels (Appendix A).

5 Conclusion

At moderate sparsity (30%), output-distribution alignment is the primary design dimension: signal type may account for most of the PPL gap (upper bound; Appendices S, T). A <1% predictor reduces TEAL's PPL by 14.5% via divergent masks; directional downstream gains but knowledge regression persist across scales.

Limitations

Our findings are bounded by three limitations. **Sparsity sweet spot.** 30% is the lowest-degradation level; at 50%, all methods collapse to near-random on reasoning tasks, and a separate predictor must be retrained per target. **Knowledge-intensive regression.** KL consistently underperforms on MMLU and GSM8K across scales (8B: -5.2pp MMLU; 14B: -21pp GSM8K), suggesting output-distribution objectives systematically sacrifice reasoning-critical pathways. **Evaluation scope.** All predictors train on WikiText-103; C4 evaluation shows preserved KL advantage but $31\times$ higher cross-seed variance. Open-ended generation (e.g., MT-Bench) and contexts beyond 2048 tokens remain untested.

References

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. 2024. On-policy distillation of language models: Learning from self-generated mistakes. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Yash Akhauri, Ahmed F AbouElhamayed, Jordan Dotzel, Zhiru Zhang, Alexander M Rush, Safeen Huda, and Mohamed S Abdelfattah. 2024. Shadowllm: Predictor-based contextual sparsity for large language models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Keivan Alizadeh, Iman Mirzadeh, Dmitry Belenko, Karen Khatamifard, Minsik Cho, Carlo C Del Mundo, Mohammad Rastegari, and Mehrdad Farajtabar. 2024. LLM in a flash: Efficient large language model inference with limited memory. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Saleh Ashkboos, Maximilian L. Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. 2024. Slicegpt: Compress large language models by deleting rows and columns. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Yoshua Bengio, Nicholas Léonard, and Aaron Courville. 2013. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*.

Runxi Cheng, Yuchen Guan, Yucheng Ding, Qingguo Hu, Yongxian Wei, Chun Yuan, Yelong Shen, Weizhu Chen, and Yeyun Gong. 2025. Mixture of neuron experts. *arXiv preprint arXiv:2510.05781*.

Nobel Dhar, Bobin Deng, Md Romyull Islam, Kazi Fahim Ahmad Nasif, Liang Zhao, and Kun Suo. 2024. Activation sparsity opportunities for compressing general large language models. *arXiv preprint arXiv:2412.12178*.

Nobel Dhar, Bobin Deng, Md Romyull Islam, Xinyue Zhang, Kazi Fahim Ahmad Nasif, and Kun Suo. 2025. A sparsity predicting approach for large language models via activation pattern clustering. In *Proceedings of the International Conference on Parallel Processing (Euro-Par)*.

Gongfan Fang, Hongxu Yin, Saurav Muralidharan, Greg Heinrich, Jeff Pool, Jan Kautz, Pavlo Molchanov, and Xinchao Wang. 2024. Maskllm: Learnable semi-structured sparsity for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Marco Federici, Davide Belli, Mart van Baalen, Amir Jalalirad, Andrii Skliar, Bence Major, Markus Nagel, and Paul Whatmough. 2024. Efficient llm inference using dynamic input pruning and cache-aware masking. In *Proceedings of Machine Learning and Systems (MLSys)*.

William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–40.

Elias Frantar and Dan Alistarh. 2023. Sparsegpt: Massive language models can be accurately pruned in one-shot. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Qichen Fu, Minsik Cho, Thomas Merth, Sachin Mehta, Mohammad Rastegari, and Mahyar Najibi. 2024. LazyLLM: Dynamic token pruning for efficient long context LLM inference. *arXiv preprint arXiv:2407.14057*.

Aayush Gautam, Mukul Gagrani, Junyoung Park, Mingu Lee, Chiris Lott, and Narasimha Reddy. 2026. Fast forward: Accelerating llm prefill with predictive ffn sparsity. *arXiv preprint arXiv:2602.00397*.

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12216–12235.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aieleen Lakber, Aimin Chen, Ait Mokhtar Salma, and Ajay Menon. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2024. Minillm: Knowledge distillation of large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Daniel Haziza, Timothy Chou, Dhruv Choudhary, Luca Wehrstedt, Francisco Massa, Jiecao Yu, Geonhwa Jeong, Supriya Rao, Patrick Labatut, and Jesse Cai. 2025. Accelerating transformer inference and training with 2:4 activation sparsity. *arXiv preprint arXiv:2503.16672*.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. In *NeurIPS 2014 Deep Learning Workshop*.

Fangcheng Huang and 1 others. 2025. LaRoSA: Accelerating large language model inference with layerwise rotated sparse activation. *arXiv preprint arXiv:2507.01299*.

Eric Jang, Shixiang Gu, and Ben Poole. 2017. Categorical reparameterization with gumbel-softmax. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, and Lucile Saulnier. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.

Samir Khaki, Xiuyu Li, Junxian Guo, Ligeng Zhu, Konstantinos N. Plataniotis, Amir Yazdanbakhsh, Kurt Keutzer, Song Han, and Zhijian Liu. 2025. Sparselora: Accelerating llm fine-tuning with contextual sparsity. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Yoon Kim and Alexander M. Rush. 2016. Sequence-level knowledge distillation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. 2024. Distillm: Towards streamlined distillation for large language models. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Donghyun Lee, Je-Yong Lee, Genghan Zhang, Mo Tiwari, and Azalia Mirhoseini. 2024. Cats: Contextually-aware thresholding for sparsity in large language models. *arXiv preprint arXiv:2404.08763*.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2024. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*.

James Liu, Pragaash Ponnusamy, Tianle Cai, Han Guo, Yoon Kim, and Ben Athiwaratkun. 2025. Training-free activation sparsity in large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Zichang Liu, Jue Wang, Tri Dao, Tianyi Zhou, Binhang Yuan, Zhao Song, Anshumali Shrivastava, Ce Zhang, Yuandong Tian, Christopher Re, and Beidi Chen. 2023. Deja vu: Contextual sparsity for efficient llms at inference time. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Christos Louizos, Max Welling, and Diederik P. Kingma. 2018. Learning sparse neural networks through l_0 regularization. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Yuqi Luo, Chenyang Song, Xu Han, Yingfa Chen, Chaojun Xiao, Xiaojun Meng, Liqun Deng, Jiansheng Wei, Zhiyuan Liu, and Maosong Sun. 2024. Sparsing law: Towards large language models with greater activation sparsity. *arXiv preprint arXiv:2411.02335*.

Chris J. Maddison, Andriy Mnih, and Yee Whye Teh. 2017. The concrete distribution: A continuous relaxation of discrete random variables. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer sentinel mixture models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Iman Mirzadeh, Keivan Alizadeh, Sachin Mehta, Carlo C Del Mundo, Oncel Tuzel, Golnoosh Samei, Mohammad Rastegari, and Mehrdad Farajtabar. 2024. Relu strikes back: Exploiting activation sparsity in large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Saurav Muralidharan, Sharath Turuvekere Sreenivas, Raviraj Joshi, Marcin Chochowski, Mostofa Patwary, Mohammad Shoeybi, Bryan Catanzaro, Jan Kautz, and Pavlo Molchanov. 2024. Compact language models via pruning and knowledge distillation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.

David Raposo, Sam Ritter, Blake Richards, Timothy Lillicrap, Peter Conway Humphreys, and Adam Santoro. 2024. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Georgii Serbin, Kirill Koshkin, Zhongao Sun, Anastasiya Bistrigova, and C.C. Korikov. 2026. Svd contextual sparsity predictors for fast llm inference. *arXiv preprint arXiv:2603.14110*.

Noam Shazeer. 2020. Glu variants improve transformer. *arXiv preprint arXiv:2002.05202*.

Jiho Shin, Hoeseok Yang, and Youngmin Yi. 2025a. GRASP: Group-based prediction of activation sparsity for fast LLM inference. In *Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC)*.

Jiho Shin, Hoeseok Yang, and Youngmin Yi. 2025b. Sparseinfer: Training-free prediction of activation sparsity for fast llm inference. In *Proceedings of the Design, Automation & Test in Europe Conference (DATE)*.

Susav Shrestha, Brian Settlemyer, Joel Dapello, and Peng Chen. 2025. Polar sparsity: High throughput batched LLM inferencing with scalable contextual sparsity. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Chenyang Song, Xu Han, Zhengyan Zhang, Shengding Hu, Xiyu Shi, Kuai Li, Chen Chen, Zhiyuan Liu, Guangli Li, Tao Yang, and Maosong Sun. 2025. Prosparse: Introducing and enhancing intrinsic activation sparsity within large language models. In *Proceedings of the International Conference on Computational Linguistics (COLING)*.

Yixin Song, Zeyu Mi, Haotong Xie, and Haibo Chen. 2024a. Powerinfer: Fast large language model serving with a consumer-grade gpu. In *Proceedings of the ACM Symposium on Operating Systems Principles (SOSP)*.

Yixin Song, Haotong Xie, Zhengyan Zhang, Bo Wen, Li Ma, Zeyu Mi, and Haibo Chen. 2024b. Turbo sparse: Achieving llm sota performance with minimal activated parameters. *arXiv preprint arXiv:2406.05955*.

Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. 2024. A simple and effective pruning approach for large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Filip Szatkowski, Bartosz Wójcik, Mikołaj Piórczyński, and Simone Scardapane. 2024. Exploiting activation sparsity with dense to dynamic-k mixture-of-experts conversion. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Rongxiang Wang, Kangyuan Shu, and Felix Xiaozhu Lin. 2025. Enabling dynamic sparsity in quantized LLM inference. *arXiv preprint arXiv:2511.04477*.

Xiuying Wei, Yunchen Zhang, Yuhang Li, Xiangguo Zhang, Ruihao Gong, Jinyang Guo, and Xianglong Liu. 2023. Outlier suppression+: Accurate quantization of large language models by equivalent and effective shifting and scaling. *arXiv preprint arXiv:2304.09145*.

Taiqiang Wu, Chaofan Tao, Jiahao Wang, Runming Yang, Zhe Zhao, and Ngai Wong. 2025a. Rethinking kullback-leibler divergence in knowledge distillation for large language models. In *Proceedings of the International Conference on Computational Linguistics (COLING)*.

Tong Wu, Yutong He, Bin Wang, and Kun Yuan. 2025b. Mixture-of-channels: Exploiting sparse FFNs for efficient LLMs pre-training and inference. *arXiv preprint arXiv:2511.09323*.

Haotian Xu, Jiannan Yang, Tian Gao, Tsui-Wei Weng, and Tengfei Ma. 2025. Resting neurons, active insights: Robustify activation sparsity for large language models. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Zhiyuan Xue, Yixin Song, Zeyu Mi, Le Chen, Yubin Xia, and Haibo Chen. 2024. Powerinfer-2: Fast large language model inference on a smartphone. *arXiv preprint arXiv:2406.06282*.

Lu Yin, You Wu, Zhenyu Zhang, Cheng-Yu Hsieh, Yaqing Wang, Yiling Jia, Gen Li, Ajay Jaiswal, Mykola Pechenizkiy, Yi Liang, Michael Bendersky, Zhangyang Wang, and Shiwei Liu. 2024. Outlier weighed layerwise sparsity (OWL): A missing secret sauce for pruning LLMs to high sparsity. In *Proceedings of the International Conference on Machine Learning (ICML)*.

Yilong Zhang, Junyuan Shang, Huanqi Cao, Shiyao Li, Shuguang Jiang, Lingxiao Ma, Shun Zheng, Cheng Li, and Jiang Bian. 2025a. COTS: Connected sparse activation for efficient LLM inference. *arXiv preprint arXiv:2502.14475*.

Zhenyu Zhang, Zechun Liu, Yuandong Tian, Harshit Khaitan, Zhangyang Wang, and Steven Li. 2025b. R-sparse: Rank-aware activation sparsity for efficient llm inference. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Zhao, Andrew M. Dai, Zhifeng Chen, Quoc V. Le, and James Laudon. 2022. Mixture-of-experts with expert choice routing. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

A Inference Efficiency

Table 5 reports inference throughput on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (96 GB; batch size 1, sequence length 2048, fp16, averaged over 100 forward passes).

Table 5: Inference latency on LLaMA-3.1-8B (RTX PRO 6000 Blackwell, batch=1, seq=2048, fp16, 100 forward passes). Overhead is relative to dense inference.

Method	Sparsity	ms/fwd	tok/s	Overhead
Dense	0%	119.11±0.57	17,195	—
KL predictor-only	30%	126.64±0.70	16,172	+6.3%
KL predictor-only	50%	126.78±0.31	16,154	+6.4%
Predictor + compensation	30%	129.94±0.75	15,761	+9.1%
Predictor + compensation	50%	129.71±0.53	15,789	+8.9%

The full KL-sparse pipeline adds 6.4% overhead to dense inference (126.78 vs. 119.11 ms/forward), of which the predictor's 32-layer forward pass contributes 3.17 ms (2.7%); the remainder arises from mask application and sparse FFN indexing. Eliminating the compensation network saves an additional ∼2.7% latency overhead and 67.2M parameters (the per-layer GEMV required for output rescaling); output-distribution KL thus provides both quality and efficiency advantages over BCE-based approaches. The overhead is near-identical at 30% and 50% sparsity because the predictor runs the same computation regardless of target sparsity; only the top-k threshold changes.

Note that the current implementation uses unstructured element-wise masks, which do not translate to wall-clock speedup without sparse kernel support. The pipeline adds ∼6.4% overhead; net acceleration requires kernel-level sparse computation optimizations (e.g., Triton block-sparse, cuSPARSE) that are beyond the scope of this work. Our contribution is prediction quality and inference simplification (removing the compensation network), not end-to-end latency reduction.

B Predictor Bottleneck Sensitivity

Table 6 reports the effect of predictor bottleneck dimension d_b on PPL at 50% sparsity under both TEAL global allocation (the paper's primary inference protocol) and per-token evaluation.

Under TEAL global allocation, increasing d_b beyond 128 provides no benefit (PPL slightly worsens), while per-token evaluation shows monotonic improvement (10.16→7.93). This divergence indicates that the global top-k allocation mechanism, not predictor capacity, is the binding constraint: additional predictor capacity produces better per-neuron scores, but the global ranking and budget distribution cannot exploit them. The d_b=128 configuration (75.9M params, <1% of base model) is therefore optimal for the TEAL inference protocol.

Table 6: Predictor bottleneck sensitivity at 50% sparsity (WikiText-2 PPL↓). TEAL global is the primary inference protocol; per-token shown for reference.

d_b	Params	TEAL global	Δ	Per-token
64	38.2M (0.48%)	12.076	+0.9%	10.16
128	75.9M (0.95%)	11.963	—	9.218
256	151.5M (1.89%)	12.105	+1.2%	8.586
512	302.5M (3.77%)	12.315	+2.9%	7.93

B.1 Temperature Schedule Sensitivity

Table 7: Temperature schedule sensitivity at 50% sparsity (WikiText-2 PPL↓, TEAL global).

τ Schedule	PPL↓
Fixed τ=0.5	**11.861**
Linear 1.0 → 0.1 (default)	11.963
Fixed τ=0.1	12.072

Temperature schedule has minimal impact (<1.8% PPL variation). The linear annealing default is a conservative choice; even removing annealing entirely (fixed τ=0.5) yields comparable performance.

B.2 Training Convergence

Table 8: Training convergence at 50% sparsity (WikiText-2 PPL↓, TEAL global).

Steps	PPL↓
200	15.158
400	12.538
600	12.046
800	11.907
1000	**11.906**

Training converges rapidly: 95% of the quality gap closes by step 600, and performance saturates by step 800 (Δ < 0.001 between 800 and 1000 steps). Our default of 1000 steps provides a conservative margin.

C Geometric Penalty Schedule Dynamics

Table 9 reports the convergence of the geometric penalty schedule across sparsity targets. The geometric doubling pattern ($1 \to 2 \to 4 \to 8 \to \ldots$) enables λ to reach steady state within ∼10 steps;

the deadzone ($\epsilon = 0.02$) then stabilizes λ once actual sparsity is within ±2% of the target.

Table 9: Geometric penalty schedule convergence across sparsity targets (margin $\epsilon = 0.02$, 1000 training steps). Deviation is |actual − target| / target.

Target ρ	$\lambda_{\max}$	Final λ	Actual sparsity	Deviation
30%	5000	5000 (cap)	0.285	5.0%
40%	5000	625	0.398	0.5%
50%	1000	8	0.504	0.8%
70%	1000	1000 (cap)	0.688	1.7%

At 50% sparsity, the SwiGLU gate distribution already clusters near the target, so λ remains low (8); the constraint is nearly satisfied by the initial predictor outputs. At 30% and 70%, the gap between the gate prior and target is larger, driving λ to its respective caps. The 40% target converges to $\lambda = 625$ with only 0.5% deviation; this precision reflects a regime where the target is reachable without λ saturation.

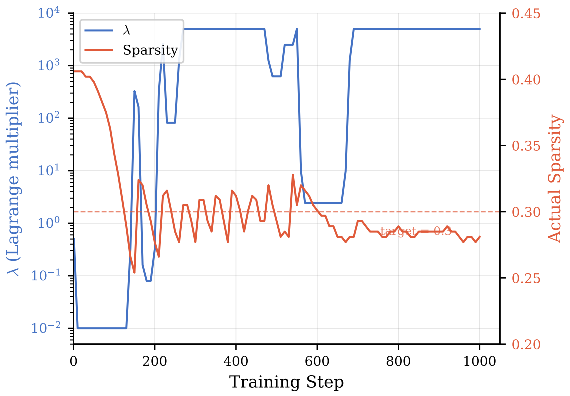

Figure 4: Lagrangian controller dynamics for 30% sparsity target ($\lambda_{\max} = 5000$). Left axis (log scale): penalty weight λ rises from 0.01 to its cap once actual sparsity undershoots the target at ∼step 150, exhibiting bang-bang oscillations before saturating. Right axis: actual sparsity (solid) converges from the natural density (≈0.41) toward the target (dashed) and stabilizes near 0.28.

Temperature τ is annealed linearly from 1.0 to 0.1 over 1000 steps. A mild loss increase in the final ∼100 steps is expected: as masks become more binary ($\tau \to 0$), discrete rounding noise slightly increases KL divergence, consistent with convergence toward the hard-mask inference regime.

The controller reliably drives actual sparsity to within 5% of the target across all tested regimes, with λ spanning three orders of magnitude (8 to 5000) via geometric doubling. For the 30% target, actual sparsity is ≈28.5% (5% deviation). This does not affect conclusions: TEAL at 28.5% actual sparsity yields PPL 10.30 (vs. 10.97 at exact 30%), a 0.67 PPL difference negligible relative to the KL vs. TEAL gap (8.96 vs. 10.97).

D Out-of-Distribution Evaluation

To address concerns about data homogeneity between training (WikiText-103) and evaluation (WikiText-2), we evaluate on C4 (Raffel et al., 2020) validation set under TEAL global allocation.

Table 10: Out-of-distribution evaluation on C4 validation (PPL↓). 30% KL reports 2-seed mean±std; other entries use seed 42.

Sparsity	Method	C4 PPL↓
0%	Dense	9.21
30%	KL predictor (Ours)	**16.03±0.01**
50%	KL predictor (Ours)	**18.40**
	TEAL vanilla	38.75

At the 30% sparsity level, the KL predictor achieves C4 PPL 16.03±0.01 (2-seed, global allocation, 1.74× dense), confirming OOD generalization. At 50%, the KL predictor achieves 52.5% relative advantage over TEAL (18.40 vs. 38.75), comparable to the 50% advantage on WikiText-2 (11.92 vs. 23.90), confirming that KL-trained predictions generalize across distributions.

Uniform Allocation. Under uniform allocation at 30%, C4 PPL across three seeds is 18.22 (s42), 16.37 (s123), 16.36 (s456), yielding mean 16.98±1.07 (1.84× dense). The cross-seed variance is substantially higher than in-domain: CV=6.3% on C4 vs. 0.2% on WikiText-2 (PPL 8.81±0.02), driven by seed 42's elevated C4 PPL (18.22 vs. 16.37 for other seeds). Global allocation's cross-layer budget flexibility partially compensates for domain shift (PPL 16.03 global vs. 16.98 uniform), consistent with the Limitations observation that the predictor partially overfits to training-domain activation patterns.

Downstream Task Robustness. Table 11 compares downstream performance when the predictor generates masks from C4 text (OOD) vs. WikiText (in-domain), both at 30% uniform allocation (seed 42). All six benchmarks show $\Delta \leq 1.1$pp, confirming that predictor masks are robust to calibration domain shift.

Table 11: Downstream task robustness to calibration domain shift (KL 30% uniform, seed 42). WikiText: predictor masks from in-domain text; C4: masks from OOD text. Metrics: acc_norm (ARC-c, HS, PIQA), acc (WG, MMLU), exact_match (GSM8K).

Calibration	ARC-c	WG	MMLU	GSM8K	HS	PIQA
WikiText	40.8	68.6	40.6	7.4	70.9	74.8
C4 (OOD)	40.4	69.3	41.4	6.3	70.7	74.0
Δ	−0.4	+0.7	+0.8	−1.1	−0.2	−0.8

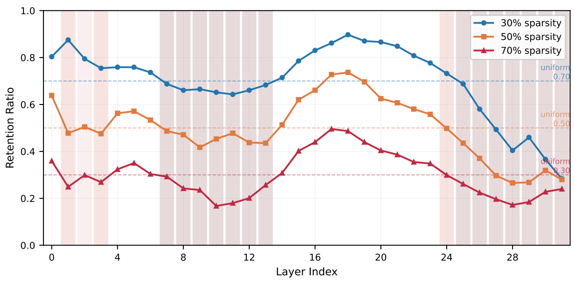

Figure 5: Per-layer neuron retention under global top-k allocation at 30%, 50%, and 70% sparsity. Dashed lines indicate uniform retention baselines. Late layers (L24–31) are consistently starved due to long-tailed activation distributions, with retention dropping to 28.4% (L31) even at 30% sparsity. At 70%, starvation extends to mid-layers (L10–12: 16.7–20.1%), explaining the transition to the capacity-limited regime.

E Per-Layer Retention Analysis

F Training Stability

Three-seed evaluation ($s \in \{42, 123, 456\}$) at 50% sparsity under uniform allocation yields PPL 11.92 ± 0.07 (CV=0.6%); training is stable across random initializations. Downstream metrics are similarly stable: ARC-c 33.8 ± 1.2%, WinoGrande 62.9 ± 0.1%, MMLU 28.0 ± 0.6%, GSM8K 2.2 ± 0.1%.

Under global allocation at 50%, MMLU is 28.3 ± 0.7%; the near-random performance is capacity-limited rather than seed-sensitive (σ=0.7pp vs. 9.5pp at 30% global). The low standard deviation across both allocation strategies indicates that the KL training landscape is well-conditioned for this predictor architecture, with low sensitivity to initialization.

F.1 Cross-Seed Mask Consistency

To assess whether KL predictors discover consistent mask patterns across initializations, we compute pairwise Jaccard distances among three independently trained KL predictors (s42/s123/s456) at 30% uniform allocation (100 sequences, WikiText-2 validation).

The mean cross-seed Jaccard distance is 0.363 (IoU = 0.637), substantially lower than the KL-vs-magnitude divergence (0.46–0.68), confirming that KL-discovered patterns form a distinct cluster from magnitude-based masks. Per-layer divergence follows a U-shaped pattern: lowest at embedding-adjacent layers (L0: 0.33), peaking at early layers (L1: 0.41), with a mid-network valley (L12–14: 0.33). The random baseline Jaccard distance is 0.462.

While specific mask configurations vary moderately across seeds (Jaccard distance 36.3%), the resulting PPL is highly stable (8.81±0.02, CV=0.2%), suggesting multiple near-equivalent optima in the activation mask space. The cross-seed divergence (36.3%) is substantially lower than the KL-vs-magnitude divergence (46–68%), confirming that KL discovers effective patterns that cluster distinctly from magnitude-based selection.

G MSE Baseline Configuration

The MSE predictor uses identical architecture (two-layer MLP, bottleneck d_b=128), training data (WikiText-103, 1000 steps), optimizer (AdamW, lr=$1{\times}10^{-3}$, weight decay $1{\times}10^{-2}$, cosine decay), Gumbel-Sigmoid mask generation (τ: $1.0 \to 0.1$), and adaptive Lagrangian sparsity control as the KL and BCE predictors. The only difference is the loss target: per-layer $\|\mathbf{W}_d^{(\ell)}(\tilde{\mathbf{m}}^{(\ell)} \odot \mathbf{g}^{(\ell)} \odot \mathbf{x}^{(\ell)} \mathbf{W}_u^{(\ell)}) - \mathbf{y}_{\text{dense}}^{(\ell)}\|_2^2$, replacing the end-to-end KL divergence. This controlled comparison isolates the effect of output-distribution alignment: all three predictors see identical data in identical order with identical architectures; only the loss function differs.

MSE uses $\lambda_{\max}$=1000 (vs. KL's 5000) because MSE's per-layer gradient magnitude is approximately $5\times$ larger than KL's end-to-end gradient, requiring proportionally less penalty pressure to achieve comparable sparsity control. Both reach their target sparsity within 110 steps. At 50% target, MSE converges to the target sparsity for seed 42; however, with seed 123, sparsity reaches only 5.1% under the same $\lambda_{\max}$ cap (eval PPL=15,789, random-level), indicating high sensitivity to initialization—a failure mode not observed with KL-based losses. At 30%, MSE achieves 27.7% actual sparsity for s42 (gap 2.3pp). The 3-seed MSE mean (18.52±1.68; s42=20.43, s123=17.73, s456=17.39) is still $1.7\times$ worse than TEAL (10.82), confirming that the performance

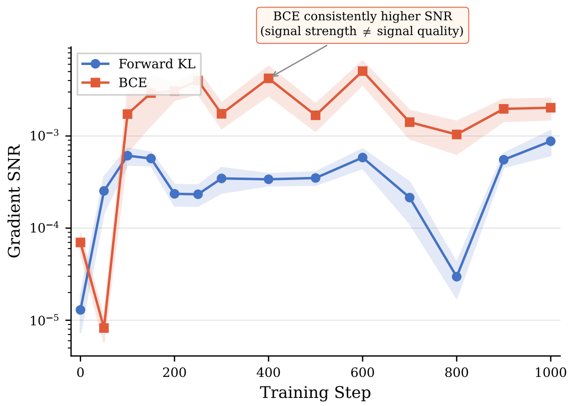

Figure 6: Gradient SNR trajectory across 1000 training steps. BCE (orange) maintains consistently higher SNR than KL (blue) from step 100 onward (2.3–17×), yet KL produces superior masks (Table 2), suggesting that target alignment matters more than gradient clarity for mask quality.

deficit stems from the loss function, not insufficient sparsity pressure.

H Extended SNR Analysis

Figure 6 plots gradient SNR across the full 1000-step training for both KL and BCE predictors, measured at 14 checkpoints (LLaMA-3.1-8B, C4) with 95% bootstrap confidence intervals. From step 100 onward, BCE maintains consistently higher SNR than KL (2.3–17× depending on training phase), with the gap narrowing toward convergence (2.3× at step 1000: 2.026×10^{-3} vs. 8.78×10^{-4}). Despite KL's noisier gradient signal, KL-trained predictors produce substantially better masks (Table 2), confirming that *target alignment* matters more than gradient clarity for mask quality. BCE's higher SNR reflects its simpler per-neuron classification objective, which concentrates gradients on individual neuron decisions; KL's coupled gradient distributes signal across all mask decisions jointly, yielding lower SNR but discovering compositionally superior mask patterns (§3.6).

I Extended Downstream Evaluation

Table 12 reports HellaSwag and PIQA across sparsity levels and allocation strategies.

At 30%, the KL predictor retains 86.3% of dense HellaSwag accuracy (71.07±0.54 vs. 82.35, 3-seed) and 92.6% of PIQA (74.95±0.25 vs. 80.90), confirming moderate degradation on non-reasoning tasks. At 50%, global allocation outperforms uniform on HellaSwag (+2.29pp), consistent with the budget-concentration benefit observed for PPL in the capacity-limited regime (§4.6); PIQA shows a smaller gap (+0.38pp).

Table 12: Extended downstream benchmarks (LLaMA-3.1-8B). HellaSwag: 10-shot normalized acc; PIQA: 0-shot normalized acc. KL 30% uniform: 3-seed mean±std; others: seed 42.

Sparsity	Method	HellaSwag	PIQA
0%	Dense	**82.35**	**80.90**
30%	KL uniform	**71.07±0.54**	**74.95±0.25**
	TEAL uniform	63.31	72.80
	BCE isocompute	62.81	74.21
50%	KL global	**58.95**	**69.48**
	KL uniform	56.66	69.10
	TEAL uniform	39.35	63.66

J Compensation Training Convergence

All compensation networks (Gautam et al., 2026) were trained for 1000 steps (lr=10^{-4}, batch size 4, seq length 2048). Training loss plateaus by step 700 across all valid runs: 30% s42 (final loss 0.70), 30% s456 (0.68), 50% s42 (1.07), with fluctuations remaining within 0.08 loss units for the final 300 steps. The compensation network converges but provides no quality improvement over the KL predictor alone, confirming that the redundancy result reflects predictor self-sufficiency rather than insufficient compensation training.

K Qwen-2.5-14B Detailed Results

Table 13: Qwen-2.5-14B at 30% nominal sparsity. Actual sparsity differs: KL 29.6% (3-seed), TEAL 29.3% (target 30%). **Bold**: best sparse per column per group. BCE collapses to 6.4% native sparsity (§3.6), so forced global top-k is used.

Method	PPL↓	ARC-c	WG	MMLU	GSM8K	HS	PIQA
Dense	5.19	67.58	81.06	79.77	87.57	84.20	81.94
Global top-k allocation							
KL (Ours)	**7.62**	57.17	72.06	63.72	48.82	**75.01**	**77.37**
TEAL	8.73	56.83	**72.69**	**71.83**	49.51	70.51	74.70
BCE	8.76	**57.59**	72.61	71.67	**53.07**	53.43	68.28
Uniform allocation							
KL (Ours)	**7.90**	53.50	**72.82**	63.55	30.9±6.8*	**72.68±0.2**	**76.01±1.1**
TEAL	8.01	**55.29**	69.85	**66.34**	**52.1**	67.72	74.70

*GSM8K: n=2 (s42/s456); s123 eval incomplete.

Under global allocation, BCE's HellaSwag drops to 53.43% (−17pp vs. TEAL), indicating that the magnitude-based predictor's mask quality degrades severely when combined with non-uniform budget distribution at 14B scale.

L Full Allocation Analysis

Table 15 extends Table 14 with all downstream metrics.

Table 14: Allocation summary (LLaMA-3.1-8B). ±: 3-seed; *high seed variance.

Sparsity	Allocation	PPL↓	MMLU
30%	Global	8.96±0.02	35.7±9.5*
	Uniform	**8.81±0.02**	**40.7±1.3**
50%	Global	**11.84±0.15**	**28.3±0.7**
	Uniform	11.92±0.07	28.0±0.6

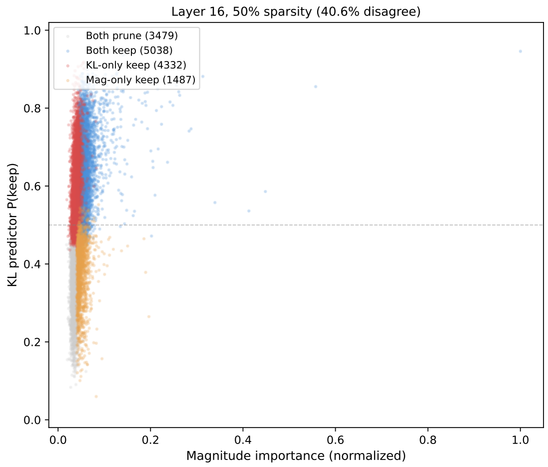

Figure 7: KL vs. magnitude importance scores for layer 16 neurons at 50% sparsity. 41% of neurons fall in disagreement quadrants where one method selects a neuron the other discards.

M Mask Pattern Visualization

Figure 8 shows per-layer Jaccard distance between KL-derived and magnitude-based masks at 30% and 50% sparsity. Figure 7 shows per-neuron importance scores at layer 16. Figure 9 shows binary mask comparisons at layers 2, 16, and 28.

N Cross-Architecture Comparison

Table 16 evaluates on Mistral-7B-v0.3 (Jiang et al., 2023) (GQA + sliding window, dense PPL 5.52).

The KL PPL advantage ratio is consistent across architectures (within 8% at $\leq$50% sparsity). Table 17 reports downstream task accuracy at 30% sparsity on Mistral-7B-v0.3 (3-seed). KL improves PPL (-17.3%) and commonsense tasks (WG +6.7pp) but MMLU is tied (43.3% vs. 43.5%); ARC-c shows high seed variance (σ=6.4pp); GSM8K reverses.

O Taxonomy Discussion

Correction granularity. *Static* methods (Xu et al., 2025) learn a fixed bias vector per layer that does not depend on the input token. *Per-token* methods (Liu et al., 2023; Gautam et al., 2026; Akhauri et al., 2024; Szatkowski et al., 2024) produce a unique mask for each input position, conditioning on the token's hidden state at each layer.

Training signal coupling. *Decomposed* objectives (BCE) treat each neuron independently, optimizing per-element classification against an oracle mask. *Intermediate* objectives (MSE on hidden states) capture local coupling within a single layer but not end-to-end output effects. *Coupled* objectives (KL on output distribution) propagate gradients through the full forward pass; each mask decision's gradient reflects its joint effect with all other masks on the final distribution.

SPON (Xu et al., 2025) distills a per-layer bias via KL but cannot adapt to per-input neuron importance (static $\times$ coupled). DejaVu (Liu et al., 2023) and FastForward (Gautam et al., 2026) achieve per-token granularity but use BCE, where each neuron's gradient is independent of other mask states. D2DMoE (Szatkowski et al., 2024) provides intermediate coupling (MSE on hidden states) but not end-to-end output effects. We define *output-distribution alignment* as whether the training loss optimizes a quantity that composes across layers. KL on output logits directly optimizes next-token prediction loss, which does not suffer from error composition; per-layer MSE minimizes reconstruction error independently at each layer, but layer-wise optima do not compose—a mask minimizing FFN output error at layer ℓ may amplify errors in layers $\ell+1 \ldots L$. The SNR analysis (§3.6) corroborates this: KL achieves the best masks despite 2.3$\times$ lower SNR than BCE, while MSE produces the worst masks. The framework predicts that losses operating on output-level quantities should outperform intermediate-level losses, a hypothesis consistent with our MSE results and testable with alternative output-level objectives.

P Oracle Mask Provenance

C1's oracle masks are extracted from a converged KL predictor and binarized via hard threshold (logits > 0, equivalent to $\sigma > 0.5$); they reflect one specific training trajectory (optimizer state, data order, Gumbel noise). The s42/s123/s456 consistency ($\leq$1.70pp) provides partial evidence of robustness, but does not rule out trajectory-dependent confounds.

Q SPON Orthogonality

SPON (Xu et al., 2025) bias correction is weakly correlated with the KL predictor (cosine similarity

Table 15: Full allocation analysis: global top-k vs. uniform across sparsity levels (LLaMA-3.1-8B). Entries with $\pm$: 3-seed mean$\pm$std ($s \in \{42, 123, 456\}$); others: seed 42. *High MMLU seed variance (individual seeds span 25.3–43.5%).

Sparsity	Allocation	PPL↓	ARC-c	WG	MMLU	GSM8K
25%	Global	**8.17**	45.4	70.6	**46.6**	**11.5**
	Uniform	8.23	**46.5**	**71.8**	44.0	11.0
30%	Global	8.96±0.02	**43.8±3.6**	**70.2±0.9**	35.7±9.5*	**8.0±0.7**
	Uniform	**8.81±0.02**	43.7±2.5	68.9±0.4	**40.7±1.3**	7.5±0.1
35%	Global	**9.31**	**38.9**	**67.6**	**40.4**	**7.7**
	Uniform	9.44	38.7	66.5	37.8	4.3
40%	Global	**10.14**	39.5	**67.5**	**39.5**	**6.6**
	Uniform	10.76	**39.6**	63.1	35.5	3.6
45%	Global	**10.80**	**39.2**	**64.6**	30.4	**3.0**
	Uniform	10.95	37.6	63.6	**30.6**	2.7
50%	Global	**11.84±0.15**	**36.3±3.2**	**63.8±0.3**	**28.3±0.7**	**2.7±1.3**
	Uniform	11.92±0.07	33.8±1.2	62.9±0.1	28.0±0.6	2.2±0.1

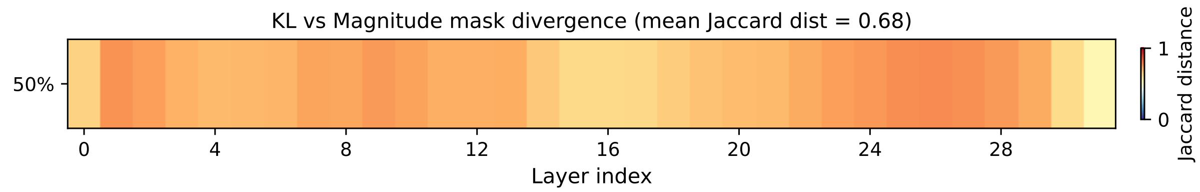

Figure 8: Per-layer Jaccard distance between KL-derived and magnitude-based masks at 30% and 50% sparsity (64 samples, 131K tokens). Divergence peaks in layers 24–28 (Jaccard distance >0.73 at 50%) and increases with sparsity across all layers.

Table 16: Cross-architecture comparison (WikiText-2 PPL↓). Adv = TEAL / KL ratio. **Note:** LLaMA results use global top-k allocation; Mistral results use uniform allocation. Mistral 30%: 3-seed mean±std; other rows: seed 42.

	Mistral-7B			LLaMA-8B		
	KL	TEAL	Adv	KL	TEAL	Adv
30%	**6.93±0.19**	8.38	1.21×	**8.96**	10.97	1.22×
50%	**8.65**	18.31	2.12×	**11.79**	23.31	1.98×
70%	**13.33**	61.72	4.63×	**17.91**	100.11	5.59×

Table 17: Mistral-7B-v0.3 downstream evaluation at 30% sparsity. $\pm$: 3-seed mean$\pm$std ($s \in \{42, 123, 456\}$); TEAL is deterministic (single run). ARC-c exhibits high seed variance (σ=6.4pp).

Method	PPL↓	ARC-c	WG	HS	PIQA	MMLU	GSM8K
Dense	5.52	60.2	78.7	83.2	81.9	62.4	36.5
KL 30%	**6.93±0.19**	**44.9±6.4**	**70.2±1.2**	**69.7±0.8**	**74.8±0.4**	43.3±1.5	8.6±2.2
TEAL 30%	8.38	41.1	63.5	60.1	72.1	**43.5**	**15.9**

0.196, R^2=0.038) and yields only 2.4% PPL improvement (12.08→11.79); the predictor already captures most recoverable signal. The decomposition confirms that SPON and the KL predictor operate on largely orthogonal error sources.

R Per-Layer KL Downstream Decomposition

Table 18 reports raw downstream numbers for the signal type vs. coupling scope decomposition at 30% sparsity (§4.5). BoolQ's ≈100% coupling is an upper bound confounded by high cross-seed variance (±11.5pp for per-layer KL); the K=4 pilot (Table 19) suggests much of this gap is train–inference mismatch.

S K=4 Mismatch Control

Table 19 reports 3-seed K=4 per-layer KL results at 30% sparsity. Training with K=4 random layers per step reduces the train–inference mismatch (4/32 vs. 1/32 layers masked during training, while inference applies all 32). The gap closure column reports the fraction of the K=1 → E2E KL gap closed by K=4.

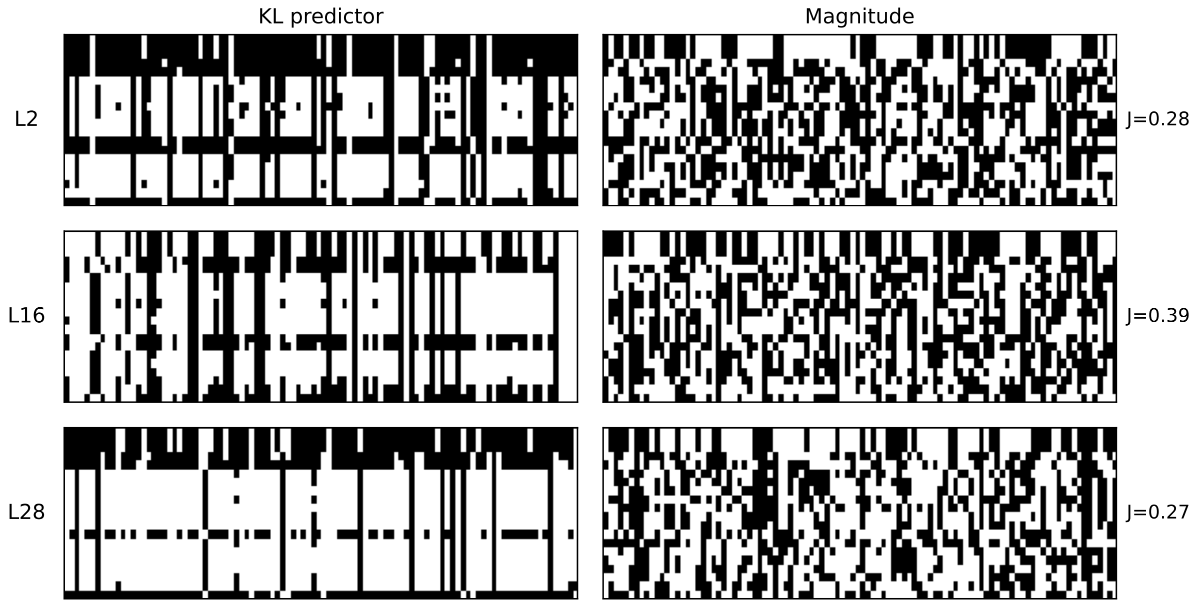

Figure 9: Binary mask comparison between KL-derived and magnitude-based masks at selected layers (L2, L16, L28) for 30% and 50% sparsity (64 samples). Black = selected by both, light regions = selected by only one method. Deep layers show the largest divergence (L28 Jaccard distance: 0.60 at 30%, 0.74 at 50%).

Table 18: 2D decomposition at 30% sparsity (LLaMA-3.1-8B, uniform allocation). All rows are 3-seed mean±std. Signal type = per-layer KL − MSE; coupling = E2E KL − per-layer KL. Percentages are upper bounds (dimensions not fully orthogonal; see §4.5).

	PPL↓	ARC-c	WG	MMLU	GSM8K	BoolQ
E2E KL	8.81±0.02	43.7	68.9	40.7	7.5	73.7
Per-layer KL	9.05±0.02	39.9	65.9	38.6	5.8	64.5
MSE	18.52±1.68	35.8±4.1	59.8±2.6	30.7±2.6	2.3±0.7	64.5±5.3
Total gap	9.71	7.9	9.1	10.0	5.2	9.2
Signal type (%)	97.5	51.9	67.0	79.0	67.3	≈0
Coupling (%)	2.5	48.1	33.0	21.0	32.7	≈100

T Per-Layer Sparsity Constraint Analysis

To test the causal role of the KL predictor's learned per-layer allocation, we evaluate two inference-time interventions at 30% overall sparsity (seed 42, uniform allocation baseline):

Protecting late layers (L22–L28) does not improve knowledge-intensive metrics despite reducing their sparsity from 48–57% to ≤30%, while worsening PPL by 5.2%. Conversely, removing the protection valley (forcing L14–L18 to 30% instead of 14–18%) improves MMLU by +3.4pp and GSM8K by +1.6pp at the cost of ARC-c −1.5pp. The KL predictor's learned allocation thus represents a task-dependent trade-off—prioritizing commonsense reasoning via middle-layer protection—rather than a uniformly optimal pattern. This allocation trade-off also explains the $K{=}4$ MMLU anomaly (§S): $K{=}4$'s near-uniform learned sparsity (std=0.003) inherently removes the valley, producing MMLU=43.5 comparable to the remove-valley condition (44.0).

Table 19: 3-seed $K{=}4$ mismatch control at 30% sparsity (LLaMA-3.1-8B, uniform allocation, $s \in \{42, 123, 456\}$). Actual sparsity 29.6%; per-layer sparsity std=0.003 (near-uniform). Gap closure $= (K{=}4 - K{=}1)/(\text{E2E} - K{=}1) \times 100$.

	$K{=}4$ (3-seed)	$K{=}1$ (3-seed)	E2E KL (3-seed)	Gap closure (%)
PPL↓	8.89±0.01	9.05±0.02	8.81±0.02	66.7
ARC-c	40.0±1.3	39.9	43.7±2.5	2.6
WG	66.9±0.2	65.9	68.9±0.4	33.3
MMLU*	42.7±1.4	38.6	40.7±1.3	>100
GSM8K	7.0±1.6	5.8	7.5±0.1	70.6
BoolQ†	67.7±7.5	64.5±11.5	73.7	34.8

*MMLU $K{=}4$ systematically exceeds E2E across all 3 seeds; attributable to $K{=}4$'s near-uniform per-layer sparsity (std=0.003) matching uniform eval allocation. Independently confirmed: removing the predictor's middle-layer valley yields MMLU=44.0% (Appendix T). †BoolQ exhibits high inter-seed variance for both $K{=}4$ (std=7.5) and $K{=}1$ (std=11.5); gap closure unreliable.

U Per-Layer Sparsity Distribution

The KL predictor at 30% target sparsity learns a non-uniform per-layer allocation that differs markedly from magnitude-based selection. Aggregate sparsity is similar across broad regions (early L0–L7: KL 28.1% vs. magnitude 27.5%; late L24–L31: 44.9% vs. 47.2%), but middle layers (L8–L23) exhibit high internal variance despite similar averages (28.2% vs. 27.3%). Table 21 highlights

Table 20: Per-layer sparsity constraint analysis. *Protect late*: cap L22–L28 sparsity $\leq$30%; *Remove valley*: override L14–L18 sparsity to 30%. Budget redistributed to maintain 30% overall.

Condition	PPL	ARC-c	MMLU	GSM8K
KL baseline (s42)	8.82	40.8	40.6	7.4
Protect late layers	9.26	42.4	40.7	6.5
Remove valley	8.76	39.3	44.0	9.0

three sub-ranges.

Table 21: Per-layer sparsity sub-ranges: KL predictor vs. magnitude (LLaMA-3.1-8B, 30%, uniform allocation). Ranges denote min–max across layers in each group.

Layer Range	KL (%)	Mag (%)	Δ (pp)
L8–L12 (high pruning)	32–36	25–26	+7 to +10
L14–L18 (protection valley)	14–18	28–30	−8 to −16
L22–L28 (aggressive pruning)	48–57	—	—

V C4 Training Data Control

To test whether WikiText-103's narrow domain drives the knowledge-task regression (§4), we train the KL predictor on C4 (Raffel et al., 2020) (diverse web text) under identical hyperparameters (1000 steps, 30% uniform, s42).

Table 22: C4-trained vs. WikiText-trained KL predictor (LLaMA-3.1-8B, 30% uniform, s42).

Training Data	PPL↓	ARC-c	WG	MMLU	GSM8K	HS / PIQA
WikiText-103	8.82	40.78	68.59	40.6	7.35	70.85 / 74.81
C4	11.55	43.09	68.67	39.95	4.09	69.24 / 75.03

The C4-trained predictor achieves comparable MMLU (40.0% vs. 40.6%) but worse GSM8K (4.1% vs. 7.4%) and higher PPL (11.55 vs. 8.82), despite training on more diverse text. This rules out training data composition as the primary cause of knowledge-task regression and supports the hypothesis that forward KL's mode-covering property structurally under-protects rare reasoning-critical activations.

W Per-Layer Coupling Breadth Analysis

We vary the number of layers coupled per training step (K) from 1 (pure per-layer) to 32 (end-to-end), evaluating all predictors under TEAL uniform allocation at 30% sparsity on LLaMA-3.1-8B.

PPL exhibits a non-monotonic U-shape (Figure 2): K=2 is worst (10.75±0.15), likely

Table 23: K-curve: coupling breadth vs. predictor quality (30%, TEAL uniform, LLaMA-3.1-8B). All values are 3-seed mean±std (s42/s123/s456) except K=1 downstream (single seed). "—" = not evaluated.

K	PPL↓	ARC-c	WG	MMLU	GSM8K	HS	PIQA
1	9.05±0.02	39.9	65.9	38.6	5.8	—	—
2	10.75±0.15	39.7±1.2	63.2±1.7	38.3±3.2	2.8±0.6	66.5±1.0	73.5±0.7
4	8.89±0.01	40.0±1.2	66.9±0.2	42.7±1.4	7.0±1.6	67.8±1.0	74.5±0.5
8	9.00±0.06	40.7±1.0	68.7±1.3	40.4±0.6	6.8±1.3	69.0±1.1	75.1±1.1
16	8.67±0.02	41.9±1.1	68.5±0.9	40.9±2.9	8.5±1.2	70.1±0.4	75.0±0.8
32 (E2E)	8.81±0.02	43.7±2.5	68.9±0.4	40.7±1.3	7.5±0.1	—	—

due to insufficient per-layer updates (∼62 updates/layer vs. ∼250 at K=8). K=16 achieves the lowest PPL (8.67±0.02), marginally better than E2E (8.81±0.02). K=16 MMLU exhibits high seed variance (std=2.9; s42=44.0, s123=40.0, s456=38.7), with the 3-seed mean (40.9) indistinguishable from E2E (40.7±1.3). The signal type gap (K=1 KL 9.05 vs. per-layer MSE 18.52±1.68) of 9.47 PPL dwarfs the coupling breadth range within KL (best K=16 8.67 vs. worst K=2 10.75 = 2.08 PPL), reinforcing that output-distribution alignment, not coupling scope, is the primary quality lever for PPL at this sparsity level.

X Licenses

LLaMA-3.1-8B is released under the Llama 3.1 Community License Agreement. Mistral-7B-v0.3 is released under the Apache 2.0 License. Qwen-2.5-14B is released under the Apache 2.0 License. WikiText-103 and WikiText-2 (Merity et al., 2017) are released under the Creative Commons Attribution-ShareAlike License (CC BY-SA 3.0). All evaluation benchmarks (ARC, WinoGrande, MMLU, GSM8K, HellaSwag, PIQA) are publicly available under their respective licenses. Our code will be released under the MIT License upon acceptance.

Y Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.